

Basic Azure Concepts

What is Microsoft Azure in simple terms?

Microsoft Azure is like renting computing services, such as servers, storage, databases, networking, and software, over the internet from Microsoft.

Instead of your company buying and managing its own physical computers (servers) in its own office (data center), you use the computers and infrastructure that Microsoft has in huge data centers all over the world. You just **pay for what you use**.

What is the difference between IaaS, PaaS, and SaaS in Azure?

These terms describe how much of the technology stack **you** manage versus how much **Microsoft (Azure)** manages for you. Think of it like deciding how much of a pizza dinner you want to prepare yourself:

Service Model	What You Manage	What Azure Manages	Analogy
IaaS (Infrastructure as a Service)	Operating System (OS), Applications, Data	Servers, Storage, Networking, Virtualization	Ordering a pizza kit. You get the raw dough, sauce, and toppings, and you bake it yourself.
PaaS (Platform as a Service)	Your Application Code and Data	OS, Servers, Storage, Networking, and all the tools needed to run code	Ordering a fresh pizza for delivery. You just eat and enjoy; you don't worry about the oven or ingredients.
SaaS (Software as a Service)	Nothing (only your data)	Everything: the whole application, including the software itself	Eating at a restaurant. You just use the completed product (like Outlook or Microsoft 365).

What is Azure App Service?

Azure App Service is a service that lets you host and run web applications, mobile app backends, and APIs **without having to manage the servers**. It's an example of a **PaaS** offering.

You just give it your code, and Azure automatically handles the server maintenance, updates, security patching, and scaling. It's a very fast and easy way to deploy a website.

What is the purpose of Azure Key Vault?

The purpose of **Azure Key Vault** is to provide a **secure place to store secrets**.

Instead of putting sensitive information like passwords, encryption keys, and API access codes directly into your application's code (which is unsafe), you store them in the Key Vault. Your application then securely asks the Key Vault for the secret only when it needs it.

Scenario-Based Beginner Questions

Your application needs to store a secret key safely. Which Azure service should you use and why?

- **Service:** Azure Key Vault
- **Why:** You should use Azure Key Vault because its primary purpose is to **centralize and securely store** sensitive data like secret keys and passwords. It uses strong encryption and tight access controls, preventing the secret from being accidentally exposed in your application's code or configuration files.

Your team wants to deploy a small web app quickly without managing servers. Which Azure service would you select?

- **Service:** Azure App Service
- **Why:** Azure App Service is a **Platform as a Service (PaaS)** offering. It lets you quickly deploy your web app code without having to spend time setting up and managing the underlying servers, operating system, or web server software. You focus only on your application code.

Your company's website is getting more traffic and slowing down. How would you scale it easily in Azure?

The easiest way is to use **Azure App Service's scaling features**:

1. **Scale Up (Vertical Scaling):** You can change the service plan to a higher tier that has more **CPU power, memory, and disk space** on the existing server.

2. **Scale Out (Horizontal Scaling):** You can easily **increase the number of server instances** (Virtual Machines) that run your application. Azure will automatically distribute the incoming traffic across all of those servers (load balancing), so no single server gets overwhelmed. This is often automated using a feature called **Autoscale**, which adds or removes servers based on traffic load.

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